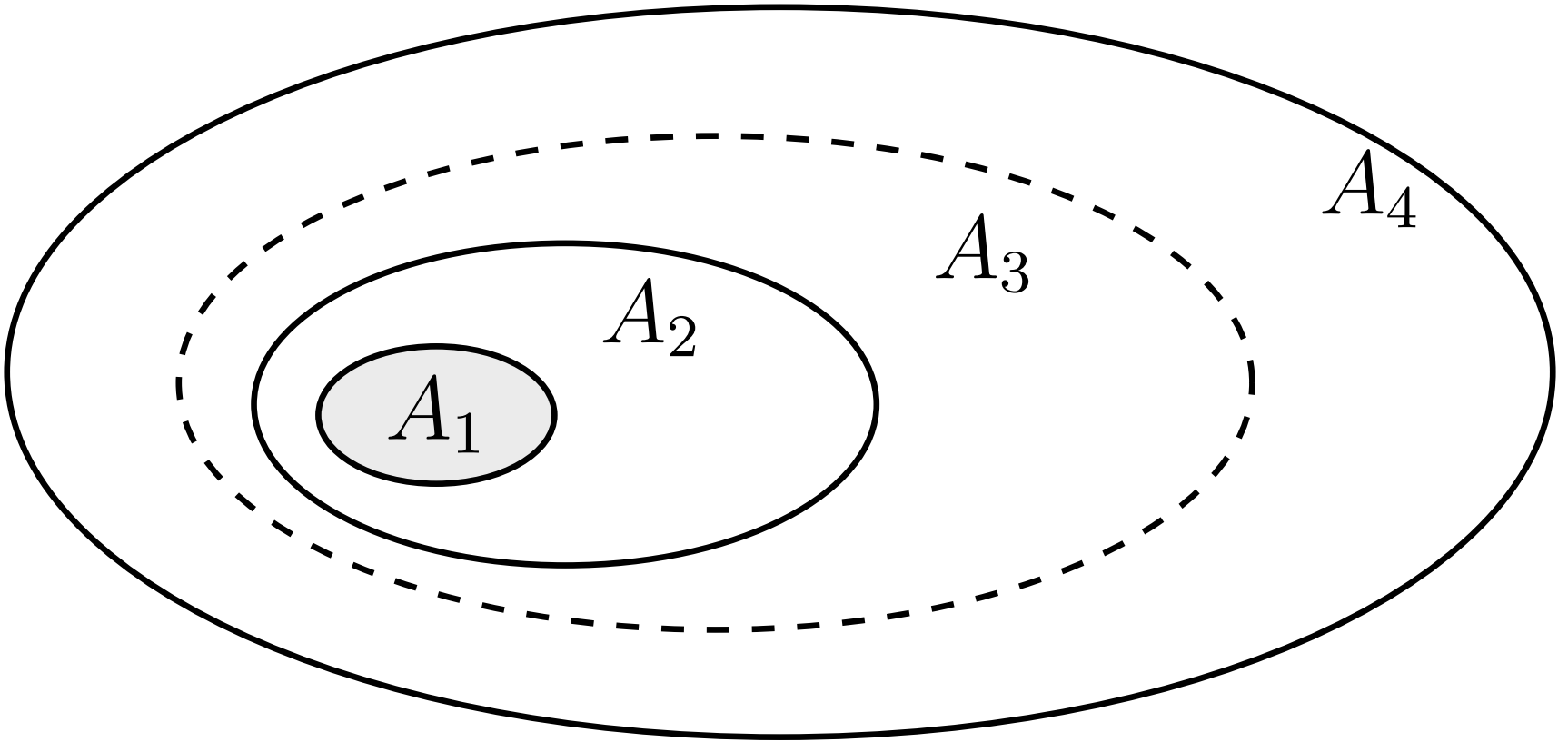

$$A_1 \subseteq A_2 \subseteq A_3 \subseteq A_4 \subseteq \cdots$$